TRIAMBAK RAMACHANDRAN

Software Developer | Machine Learning Engineer
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PROFESSIONAL SUMMARY

Senior Data Engineer at Genpact, promoted within 1.5 years, owning a Databricks Lakehouse platform serving 10K+ users across 40+ business domains. Builds Medallion-architecture pipelines in Python, PySpark and SQL with Delta Live Tables, delivering zero-downtime migrations with 40–60% faster pipelines and 25% lower compute cost. Applies machine learning, deep learning, NLP and Generative AI — spanning neural architecture search, model compression, segmentation and recommendation systems. Databricks Certified Data Engineer Professional.

PROFESSIONAL EXPERIENCE

Genpact | Bengaluru, India

Senior Data Engineer | Oct 2024 – Present

- Own technical strategy and delivery for a large-scale HR data platform on Databricks serving **10K+ users across 40+ business domains**; promoted within 1.5 years.
- Architect **PySpark** and **Delta Live Tables** pipelines across the full Medallion stack (Bronze / Silver / Gold), enforcing data quality checks, SLAs and schema governance.
- Led **zero-downtime** UKG → Workday HR data migration for 10K+ employees; implemented **Auto Loader** with CDC merge for streaming ingestion at sub-5-minute latency.
- Migrated enterprise datasets from **Greenplum / PostgreSQL** to the Lakehouse and replaced legacy Talend jobs with **Python**-based **Airflow** DAGs, adding orchestration, SLA enforcement, retries and alerting.
- Optimized **PySpark** joins, partitioning and shuffles and tuned **SQL** warehouses via Z-Ordering, result caching and materialized views — **40% faster pipelines, 25% lower compute cost, 60% less manual reconciliation**.
- Built CI/CD with **Databricks Asset Bundles** and Azure DevOps; drive alignment on observability, monitoring and incident response.

Genpact | Hyderabad, India

Data Science Intern | Feb 2024 – Aug 2024

- Built a **PDF question-answering assistant** using **retrieval-augmented generation (RAG)** — document chunking, vector embeddings and semantic search — served via the **OpenAI API** alongside custom **Hugging Face** transformer models.
- Built customer segmentation models for retail personalization using clustering (K-Means, hierarchical) and collaborative-filtering recommendation systems.
- Applied NLP to customer feedback and review text — tokenization, embeddings and sentiment scoring — to enrich segmentation features.
- Ran statistical analysis and visualization on engagement data to optimize personalization strategies for business stakeholders.

MulticoreWare | Remote

Research Intern — Machine Learning | Jan 2023 – Dec 2023

- Implemented **neural architecture search** pipelines with **NSGA-Net** and **NEAT**, using multi-objective evolutionary optimization to co-optimize accuracy, parameter count and inference latency.
- Extended the **EvoPrunerPool** framework with genetic-algorithm-driven filter selection, compressing CNNs into compact deployment-ready models that retained baseline accuracy.
- Automated hyperparameter tuning with evolutionary strategies (population-based mutation and crossover), replacing manual grid search and converging faster on optimal configurations.
- Prototyped **Differentiable Architecture Search (DARTS)** for gradient-based cell discovery; benchmarked all NAS variants on standard vision datasets for accuracy vs. FLOPs and wall-clock efficiency.

TECHNICAL SKILLS

Languages: Python, **SQL**, **PySpark**, Java, C++, C, JavaScript

Data Engineering: Databricks, Apache Spark, PySpark, Delta Lake, Delta Live Tables (DLT), Lakeflow Connect, dbt, Apache Airflow, Talend, ETL / ELT, batch and streaming pipelines

Modeling & Optimization: Medallion / Lakehouse architecture, data modeling, star schema, fact and dimension tables, SCD Type 2, Z-Ordering, partitioning, query profiling

Cloud & Storage: AWS S3, Azure Data Lake, Unity Catalog, PostgreSQL, MySQL, Oracle, Greenplum

ML & AI: TensorFlow, PyTorch, Keras, scikit-learn, deep learning, 3D CNNs, Vision Transformers, OpenCV, NAS (NSGA-Net, NEAT, DARTS), evolutionary algorithms, model pruning and compression, clustering, recommendation systems, NLP, Generative AI / LLMs, RAG, vector embeddings, Hugging Face, OpenAI API

DevOps & Practices: Git, Docker, Azure DevOps, Databricks Asset Bundles, CI/CD, Agile delivery, data quality, monitoring and alerting

PROJECTS

Violence Detection | Python, TensorFlow, Keras, 3D CNNs, Vision Transformers | Jan 2024 – May 2024

- Developed a video-based violence detection system using Vision Transformers, X3D, MobileNet with attention and a lightweight 3D CNN.
- Trained and benchmarked models on the **RWF-2000** and **RLVS** datasets, comparing architectures and tuning hyperparameters to maximize accuracy and inference efficiency.

Faculty Dashboard | HTML, CSS, JavaScript, Flask, MySQL | Feb 2023 – Jun 2023

- Built a faculty management system with attendance tracking, grade management and assignment submission, automating administrative workflows and surfacing performance trends for staff.
- Ensured reliability through unit, integration and end-to-end testing with JUnit, Postman and Selenium.

EDUCATION

B.Tech, Computer Science & Engineering | Amrita Vishwa Vidyapeetham, Coimbatore, India | 2020 – 2024 | CGPA: 8.65 / 10

CERTIFICATIONS

Databricks Certified Data Engineer Professional · Databricks Certified Data Engineer Associate